

Signed low-entropy reweightings to rank one on the sphere

A campaign report: Question 8.1 of Barak–Kothari–Steurer, answered

AICR campaign c/0048 · artifact 0048-RKHS-1-r5

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Result. Question 8.1 of Barak–Kothari–Steurer (arXiv:1701.06321v2, p. 19) has an affirmative answer, and a much stronger one than it asks for: entropy cost $\log(n(n+1)/2) = O(\log n)$, Frobenius error exactly 0, and a constant independent of ε . The paper asks whether cost $O(n^\delta)$ is achievable for arbitrarily small $\delta > 0$; $O(\log n)$ is below every n^δ .

Verification status. Not certified. Four rounds of five independent blind referee passes (20 passes, 3 models, 5 angles) have been run. No pass in any round has faulted a load-bearing step; every substantive defect found has been in the scope commentary, and the load-bearing text is byte-identical across all five revisions. Best round scored 4 clean of 5; the current revision r5 is unverified. Details in §7.

Caveat that matters. The passes were run as in-session subagents, not as the isolated `--bare` processes the harness specifies, because this host has no model backend. One vendor, three models. See §8.

1 The question

The source paper’s Question 8.1 reads, verbatim (p. 19):

Is it the case that for every distribution μ over $\mathbb{S}^{n-1}$ and every $\varepsilon, \delta > 0$ there is a (not necessarily positive) function $r : \mathbb{S}^{n-1} \rightarrow \mathbb{R}$ such that $\mathbb{E}_{v \sim \mu} |r(v)| = 1$, $\mathbb{E}_{v \sim \mu} |r(v)| \log |r(v)| \leq O(n^\delta)$ and a nonzero rank one L such that $\|\mathbb{E}_{v \sim \mu} [r(v)vv^\top] - L\|_F \leq \varepsilon \|L\|_F$?

The paper adds: “A positive solution for Question 8.1 for any $\delta < 1/2$ would be very interesting. It may improve the best known bound for the log rank conjecture to $\tilde{O}(n^\delta)$ and if appropriately extended to pseudo-distributions, improve our algorithm’s running time to $\exp(\tilde{O}(n^\delta))$ as well. We do know that the answer to this question is No if one does not allow negative reweighting functions.”

The formal statement attacked here was transcribed independently from that PDF and then checked against it; the transcription is faithful, with no additional hypothesis. Call a Borel measurable, μ -integrable $r : \mathbb{S}^{n-1} \rightarrow \mathbb{R}$ — not required nonnegative — a *signed reweighting of μ with entropy cost at most K* if

$$\mathbb{E}_{v \sim \mu} |r(v)| = 1, \quad \mathbb{E}_{v \sim \mu} [|r(v)| \log |r(v)|] \leq K, \quad \text{convention } 0 \log 0 := 0.$$

Where the difficulty was expected to lie. Write $|r| = w$ and $\sigma = \text{sign } r$. Then $\mathbb{E}_\mu |r| = 1$ makes $d\nu = w d\mu$ a probability measure with $\Delta_{\text{KL}}(\nu \| \mu) = \mathbb{E}_\mu [w \log w]$, and $\mathbb{E}_\mu [r vv^\top] = \mathbb{E}_\nu [\sigma(v) vv^\top]$. So the question asks: tilt μ at entropy price K , then choose signs freely, and make the resulting second moment relatively close to rank one. In the nonnegative case $\sigma \equiv 1$ the matrix $\mathbb{E}_\nu [vv^\top]$ is PSD of trace 1, so $\|\mathbb{E}_\nu [vv^\top]\|_F \geq n^{-1/2}$: the target cannot be shrunk, and that floor is what makes the nonnegative problem cost $\sqrt{n}$. Signs remove the floor. The proof below removes it completely — it makes the second moment *exactly* a rank one matrix.

2 The result

Theorem 1. *For every $n \geq 1$ and every Borel probability measure μ on $\mathbb{S}^{n-1}$ there exist a point $v_0 \in \text{supp } \mu$, a constant c with $(n(n+1)/2)^{-1/2} \leq c \leq 1$, and a signed reweighting r of μ — indeed the restriction to $\mathbb{S}^{n-1}$ of a single quadratic form, hence a bounded Borel function — such that*

$$\mathbb{E}_{v \sim \mu} [r(v) v v^\top] = c v_0 v_0^\top \quad \text{exactly,}$$

and whose entropy cost satisfies

$$\mathbb{E}_\mu [|r| \log |r|] \leq \log K(v_0, v_0) \leq \log(n(n+1)/2).$$

Here $L := c v_0 v_0^\top$ is rank one, nonzero and PSD, with $\|L\|_F = c$.

Corollary 2. *Question 8.1 holds with $C(\varepsilon, \delta) := 2/(\varepsilon\delta)$, a constant independent of ε : for every $\varepsilon > 0$, $\delta > 0$, every n and every Borel probability μ on $\mathbb{S}^{n-1}$ there are a signed reweighting of entropy cost at most $\log(n(n+1)/2) \leq (2/(\varepsilon\delta)) n^\delta$ and a nonzero rank one L with $\|\mathbb{E}_\mu[r(v) v v^\top] - L\|_F = 0 \leq \varepsilon \|L\|_F$.*

The mechanism, in one sentence. On the sphere the constant function 1 is a quadratic form, namely $v \mapsto v^\top I v = \|v\|_2^2$; so the reproducing kernel of the quadratic forms inside $L^2(\mu)$ has mean exactly 1, which normalises it for free, and it reproduces every coordinate product $v_a v_b$, which makes its second moment exactly $v_0 v_0^\top$. The entropy price is then the logarithm of its squared L^2 norm, which averages over μ to the dimension of the space, at most $n(n+1)/2$.

3 The proof

Throughout, $\mathbb{S}^{n-1}$ carries its usual compact, second-countable metric topology; μ is a Borel probability measure on it; Sym_n denotes the real symmetric $n \times n$ matrices; and for $Q \in \text{Sym}_n$ we write $q_Q(v) = v^\top Q v$. Two facts about the support are used and recorded now.

(P1) $\mu(\text{supp } \mu) = 1$. The complement of the support is a union of open null sets; since $\mathbb{S}^{n-1}$ is second countable that union is a countable union of null sets, hence null.

(P2) If $v \in \text{supp } \mu$ and $U \ni v$ is open, then $\mu(U) > 0$. This is the definition of the support.

Lemma 3 (the quadratic space, evaluation on the support, and its kernel). *Let $F_\mu := \{[q_Q] \in L^2(\mu) : Q \in \text{Sym}_n\}$, where $[\cdot]$ is the μ -a.e. class, and let $d := \dim F_\mu$. Then*

- (a) F_μ is a linear subspace of $L^\infty(\mu) \cap L^2(\mu)$ with $d \leq n(n+1)/2$ and $\|q_Q\|_\infty \leq \|Q\|_F$;
- (b) if $q_Q = q_{Q'}$ holds μ -a.e. then $q_Q(v) = q_{Q'}(v)$ for every $v \in \text{supp } \mu$; hence for $v \in \text{supp } \mu$ the map $\text{ev}_v : F_\mu \rightarrow \mathbb{R}$, $[q_Q] \mapsto q_Q(v)$, is well defined and linear;
- (c) for each $v \in \text{supp } \mu$ there is a unique $k_v \in F_\mu$ with $\langle k_v, f \rangle_{L^2(\mu)} = \text{ev}_v(f)$ for all $f \in F_\mu$; set $K(v, v) := \|k_v\|_{L^2(\mu)}^2 = \text{ev}_v(k_v)$;
- (d) $[1] = [q_I] \in F_\mu$, and $[v \mapsto v_a v_b] = [q_{E_{ab}}] \in F_\mu$ where $E_{ab} = (e_a e_b^\top + e_b e_a^\top)/2$.

Proof. (a) $Q \mapsto q_Q$ is linear, so F_μ is the image of Sym_n and $d \leq \dim \text{Sym}_n = n(n+1)/2$. For $v \in \mathbb{S}^{n-1}$, $|v^\top Q v| \leq \|Q\|_{\text{op}} \|v\|_2^2 = \|Q\|_{\text{op}} \leq \|Q\|_F$. Since μ is a probability measure, $L^\infty(\mu) \subseteq L^2(\mu)$.

(b) Put $h := q_Q - q_{Q'}$, a polynomial and hence continuous, with $\mu(\{h \neq 0\}) = 0$. Suppose $h(v_0) \neq 0$ for some $v_0 \in \text{supp } \mu$. Then $U := \{|h| > |h(v_0)|/2\}$ is open and contains v_0 , so $\mu(U) > 0$ by (P2); but $U \subseteq \{h \neq 0\}$ is null — a contradiction. Hence $h \equiv 0$ on $\text{supp } \mu$. Well-definedness of ev_v follows: ev_v is defined through quadratic-form representatives only, and

by the displayed statement any two quadratic-form representatives of the same class of F_μ take the same value at every $v \in \text{supp } \mu$; linearity is inherited from $Q \mapsto q_Q(v)$.

(c) F_μ is a finite-dimensional subspace of $L^2(\mu)$, hence complete, hence a Hilbert space; ev_v is a linear functional on it, hence bounded, so Riesz representation supplies a unique $k_v \in F_\mu$. Explicitly $k_v = \sum_j \text{ev}_v(f_j) f_j$ for any orthonormal basis (f_j) . Taking $f = k_v$ gives $K(v, v) = \text{ev}_v(k_v)$.

(d) $q_I(v) = \|v\|_2^2 = 1$ identically on $\mathbb{S}^{n-1}$, and $q_{E_{ab}}(v) = v_a v_b$ identically. $\square$

Part (b) is the only place where the general Borel case differs from the finitely supported one, and it is what removes the need for any approximation argument: for finitely supported μ it is trivial, every support point being an atom.

Lemma 4 (three reproducing identities). *Let $v_0 \in \text{supp } \mu$ and $k := k_{v_0}$. Then*

$$(i) \mathbb{E}_\mu[k(v) v v^\top] = v_0 v_0^\top;$$

$$(ii) \mathbb{E}_\mu[k] = 1; \text{ hence } \mathbb{E}_\mu|k| \geq 1, K(v_0, v_0) \geq 1, \text{ and } k \neq 0;$$

$$(iii) \mathbb{E}_\mu[k^2] = K(v_0, v_0) \text{ and } 1 \leq \mathbb{E}_\mu|k| \leq K(v_0, v_0)^{1/2} < \infty.$$

Proof. All integrals converge absolutely: k is bounded by Lemma 3(a), and the other factors are bounded by 1 on $\mathbb{S}^{n-1}$.

(i) Fix a, b . By Lemma 3(d) the class $g_{ab} := [v \mapsto v_a v_b]$ lies in F_μ , so the reproducing property gives $\mathbb{E}_\mu[k v_a v_b] = \langle k, g_{ab} \rangle = \text{ev}_{v_0}(g_{ab}) = (v_0)_a (v_0)_b$. The (a, b) entries of the two matrices agree, for all a, b .

(ii) By Lemma 3(d), $[1] \in F_\mu$ with $\text{ev}_{v_0}([1]) = q_I(v_0) = 1$, so $\mathbb{E}_\mu[k] = \langle k, [1] \rangle = 1$. Then $\mathbb{E}_\mu|k| \geq |\mathbb{E}_\mu k| = 1$ and, by Cauchy–Schwarz, $K(v_0, v_0) = \mathbb{E}_\mu[k^2] \geq (\mathbb{E}_\mu|k|)^2 \geq 1$. And $\mathbb{E}_\mu[k] = 1 \neq 0$ forces $k \neq 0$.

(iii) $K(v_0, v_0) = \|k\|^2 = \mathbb{E}_\mu[k^2]$ by definition; finiteness is Lemma 3(a); the upper bound is Cauchy–Schwarz against 1. $\square$

Statement (ii) is precisely the trace of (i), since $\text{tr } \mathbb{E}_\mu[k v v^\top] = \mathbb{E}_\mu[k \|v\|_2^2] = \mathbb{E}_\mu[k]$ while $\text{tr}(v_0 v_0^\top) = 1$. This locates where the construction gets its normalisation for free: the sphere constraint $\|v\|_2 = 1$ is what makes the constant function a quadratic form. On a domain where $\|v\|_2$ were not constant, neither $[1] \in F_\mu$ nor $\mathbb{E}_\mu[k] = 1$ would be available.

Lemma 5 (a cheap base point exists). $\int K(v, v) d\mu(v) = d$, and consequently there exists $v_0 \in \text{supp } \mu$ with $K(v_0, v_0) \leq d \leq n(n+1)/2$.

Proof. Fix an orthonormal basis $f_1, \dots, f_d$ of F_μ and quadratic-form representatives q_j of f_j . For $v \in \text{supp } \mu$, Lemma 3(c) and Parseval give $k_v = \sum_j q_j(v) f_j$, hence

$$K(v, v) = \sum_{j=1}^d q_j(v)^2 =: g(v) \quad \text{for all } v \in \text{supp } \mu. \quad (1)$$

Now g is a polynomial, hence Borel, and by (P1) it represents $v \mapsto K(v, v)$ up to a null set, so the integral is well defined and

$$\int K(v, v) d\mu = \sum_j \int q_j^2 d\mu = \sum_j \|f_j\|^2 = d. \quad (2)$$

(For $v \in \text{supp } \mu$ the value $g(v) = \|k_v\|^2$ is independent of the choice of basis and of representatives, by uniqueness in Lemma 3(c); off the support nothing below uses g .)

Let $A := \{g \leq d\}$, a Borel set. If $\mu(A) = 0$ then $h := g - d$ satisfies $h > 0$ μ -a.e. while $\int h d\mu = 0$ by (2); a nonnegative function with vanishing integral vanishes a.e., so $h = 0$ μ -a.e., which cannot hold simultaneously with $h > 0$ μ -a.e. under a probability measure. So $\mu(A) > 0$, hence by (P1) $A \cap \text{supp } \mu \neq \emptyset$, and any v_0 in it satisfies $K(v_0, v_0) = g(v_0) \leq d$. $\square$

Identity (2) is the statement $\int K(v, v) d\mu = \text{tr } P$ for P the orthogonal projection of $L^2(\mu)$ onto F_μ — equivalently, that the reciprocal Christoffel function has μ -mean d . It is proved above from scratch and depends on no external source.

Lemma 6 (entropy at most the log of the second moment). *If $r \in L^2(\mu)$ and $\mathbb{E}_\mu|r| = 1$, then $\mathbb{E}_\mu[|r| \log |r|] \leq \log \mathbb{E}_\mu[r^2]$.*

Proof. Write $\varphi(t) = t \log t$ for $t > 0$ and $\varphi(0) = 0$. Then $\varphi(t) \geq -1/e$ for $t \geq 0$, and $\varphi(t) \leq t^2$ for all $t \geq 0$ (for $t \geq 1$ because $\log t \leq t$; for $t < 1$ because $\varphi \leq 0$). Since $r \in L^2(\mu)$ and μ is a probability measure, $\varphi(|r|) \in L^1(\mu)$, so the entropy is a finite real number.

Let $d\nu := |r| d\mu$. Then $\nu(\mathbb{S}^{n-1}) = \mathbb{E}_\mu|r| = 1$, so ν is a probability measure, and $\nu(\{r = 0\}) = 0$, so $\log |r|$ is ν -a.e. finite and $\mathbb{E}_\nu[\log |r|] = \mathbb{E}_\mu[|r| \log |r|]$, the convention $0 \log 0 = 0$ matching the ν -null set $\{r = 0\}$. Also $\mathbb{E}_\nu[r] = \mathbb{E}_\mu[r^2] < \infty$. Jensen's inequality for the concave function $\log$ against ν gives $\mathbb{E}_\nu[\log |r|] \leq \log \mathbb{E}_\nu[r] = \log \mathbb{E}_\mu[r^2]$. $\square$

Equality holds exactly when $|r|$ is ν -a.s. constant, i.e. when $|r| \in \{0, c\}$ μ -a.e., in which case the bound reads $\log(1/\mu(\{|r| = c\}))$. So Lemma 6 is tight precisely on the indicator-type reweightings — those for which “reweighting” means conditioning on an event — and is lossy exactly to the extent that $|r|$ is spread out.

Proof of Theorem 1

Let n and μ be given. Lemma 5 supplies $v_0 \in \text{supp } \mu$ with $K(v_0, v_0) \leq d \leq n(n+1)/2$; put $k := k_{v_0}$. By Lemma 4(ii),(iii) we have $1 \leq \mathbb{E}_\mu|k| \leq K(v_0, v_0)^{1/2} < \infty$ and $k \neq 0$, so

$$c := \frac{1}{\mathbb{E}_\mu|k|} \in [K(v_0, v_0)^{-1/2}, 1] \subseteq [(n(n+1)/2)^{-1/2}, 1]$$

is well defined and strictly positive. Put $r := ck$, with representative the quadratic form q_{cQ_k} ; then r is a bounded Borel function on $\mathbb{S}^{n-1}$, hence μ -integrable and in $L^2(\mu)$, and $\mathbb{E}_\mu|r| = c \mathbb{E}_\mu|k| = 1$. So r is an admissible signed reweighting.

Second moment. By Lemma 4(i), $\mathbb{E}_\mu[r(v)vv^\top] = cv_0v_0^\top =: L$. Since $c > 0$ and $\|v_0\|_2 = 1$, $L = (cv_0)v_0^\top$ is rank one, nonzero and PSD with $\|L\|_F = c > 0$. The Frobenius error is $0 \leq \varepsilon\|L\|_F$ for every $\varepsilon > 0$.

Entropy. Lemma 6 applies, and with Lemma 4(iii),

$$\begin{aligned} \mathbb{E}_\mu[|r| \log |r|] &\leq \log \mathbb{E}_\mu[r^2] = \log(c^2 K(v_0, v_0)) = \log \frac{K(v_0, v_0)}{(\mathbb{E}_\mu|k|)^2} \\ &\leq \log K(v_0, v_0) \leq \log \frac{n(n+1)}{2}, \end{aligned}$$

using $\mathbb{E}_\mu|k| \geq 1$ from Lemma 4(ii) and Lemma 5. $\square$

Proof of Corollary 2

For $n = 1$, $\log(n(n+1)/2) = \log 1 = 0$. For $n \geq 2$, $n(n+1)/2 \leq n^2$ gives $\log(n(n+1)/2) \leq 2 \log n$; and $\max_{x>0}(\log x)/x^\delta = 1/(e\delta)$, attained at $x = e^{1/\delta}$, gives $2 \log n \leq (2/(e\delta))n^\delta$. Combine with Theorem 1. $\square$

4 Two supporting statements

Remark 7 (the conclusion is not obtainable by cancelling to zero). Let $M \in \text{Sym}_n$ have eigenvalues indexed so that $|\lambda_1| \geq |\lambda_2| \geq \dots$, and let $\varepsilon \in (0, 1)$. If $M = 0$, no admissible L exists, since $\|0 - L\|_F = \|L\|_F > \varepsilon\|L\|_F$ for every $L \neq 0$. If $M \neq 0$, a nonzero rank one L with $\|M - L\|_F \leq \varepsilon\|L\|_F$ exists *if and only if* $\sum_{i \geq 2} \lambda_i^2 \leq (\varepsilon^2/(1 - \varepsilon^2))\lambda_1^2$.

Proof. Write $L = t u w^\top$ with $\|u\| = \|w\| = 1$ and $t > 0$, so $\|L\|_F = t$ and $\|M - L\|_F^2 = \|M\|_F^2 - 2t u^\top M w + t^2$. The requirement is $(1 - \varepsilon^2)t^2 - 2(u^\top M w)t + \|M\|_F^2 \leq 0$, and increasing $u^\top M w$ only helps, so the optimal choice is $u^\top M w = s_1(M) = |\lambda_1|$ by the SVD. The quadratic in t has a real root iff $s_1^2 \geq (1 - \varepsilon^2)\|M\|_F^2 = (1 - \varepsilon^2)\sum_i \lambda_i^2$, which rearranges to the stated inequality; and when $M \neq 0$ both roots are positive, their product being $\|M\|_F^2/(1 - \varepsilon^2) > 0$ and their sum $2s_1/(1 - \varepsilon^2) > 0$, so an admissible $t > 0$ exists exactly then. $\square$

This is not load-bearing. It is recorded because it shows that the relative normalisation $\|\cdot\|_F \leq \varepsilon\|L\|_F$ does real work: the conclusion cannot be reached by cancelling the second moment to zero, so Theorem 1 is not exploiting a degenerate reading of the question.

Corollary 8 (the same for rank one matrices of unit Frobenius norm). *Let $\mathcal{X} := \{X \in \mathbb{R}^{n \times n} : \text{rank } X = 1, \|X\|_F = 1\}$, equivalently $X = ab^\top$ with $\|a\|_2 = \|b\|_2 = 1$ — the normalisation used in the source paper’s intended application. For every Borel probability measure μ on $\mathcal{X}$ there exist $X_0 \in \text{supp } \mu$, $c \in (0, 1]$ and a signed reweighting r of μ with $\mathbb{E}_\mu[|r| \log |r|] \leq \log(n^2 + 1)$ such that $\mathbb{E}_{X \sim \mu}[r(X) X] = c X_0$, exactly rank one and nonzero.*

Proof. Run Lemmas 3–6 verbatim with $\mathbb{S}^{n-1}$ replaced by the compact set $\mathcal{X}$ and the quadratic forms replaced by the affine space $G := \{X \mapsto \alpha + \langle A, X \rangle : \alpha \in \mathbb{R}, A \in \mathbb{R}^{n \times n}\}$, of dimension at most $n^2 + 1$. Only two properties of the function space were ever used: it contains the constant 1 (here trivially, $\alpha = 1, A = 0$) and it contains each coordinate function $X \mapsto X_{ab}$ (here $\alpha = 0, A = e_a e_b^\top$). Continuity of the elements of G gives Lemma 3(b) unchanged, and boundedness on $\mathcal{X}$ gives Lemma 3(a). $\square$

So the mechanism is not an artefact of specialising to $X = vv^\top$: what it needs is the presence of constants and coordinates in one finite-dimensional function space. On the sphere the homogeneous quadratic forms suffice because $\|v\|_2 = 1$ makes the constant a quadratic form, and that buys only the sharper bound $\log(n(n+1)/2)$ over $\log(n^2 + 1)$.

5 Scope: what this does and does not establish

Theorem 1 answers Question 8.1 as literally posed, and nothing more. Three delimitations, each checked against the source text.

(i) No conflict with the source’s negative results. The remark that “*as stated, Theorem 2.3 is tight*” concerns *deficient* reweightings, which by the paper’s Definition 2.2 are probability distributions and hence nonnegative; and “*the answer to this question is No if one does not allow negative reweighting functions*” is an existential statement over μ . Neither is contradicted here. Whenever the construction returns a nonnegative r , that r is itself a nonnegative reweighting of cost at most $\log(n(n+1)/2) = O(\log n)$ — more precisely, $k \geq 0$ forces $\mu(\{\pm v_0\}) = 1/K(v_0, v_0) \geq 2/(n(n+1))$ — so such a μ is not a hard instance for the nonnegative question. The same holds on the domain of Corollary 8, where the corresponding bound is the single-atom $\mu(\{X_0\}) = 1/K(X_0, X_0) \geq 1/(n^2 + 1)$. What is *not* addressed is the unnormalised domain of the paper’s Theorem 2.3, “any distribution over rank one $n \times n$ matrices”; Corollary 8 treats only the unit-Frobenius slice, and nothing is claimed beyond it.

(ii) No log rank consequence is claimed. The paper’s dictionary from reweightings to submatrices is stated only for *flat* reweightings — “*a flat distribution μ' with $\Delta_{\text{KL}}(\mu' \parallel \mu) \leq k$ corresponds to the uniform distribution over $\{u_i v_i^\top\}_{i \in I}$ where $I \subseteq [N]$ satisfies $|I| \geq 2^{-k} N$* ” — and Theorem 1 supplies no flatness guarantee, so that route is unavailable. No other route is addressed, and the paper’s dual Theorem 2.4 is untouched. Morally the target $c v_0 v_0^\top$ is the $|I| = 1$ answer; the content of Theorem 1 is that cancellation reaches *that same matrix* at cost $\log(n(n+1)/2)$, where conditioning would have cost $\log(1/\mu(\{v_0\}))$ — infinite for continuous μ .

(iii) **The pseudo-distribution version is out of scope, and the obstruction is identifiable.** The inner product used, $\langle q_Q, q_{Q'} \rangle = \mathbb{E}_\mu[(v^\top Q v)(v^\top Q' v)]$, is a degree-4 moment and does have a pseudo-distribution analogue. What has none is ev_{v_0} : a pseudo-distribution has no support points, and Lemma 3(b) — the step that makes point evaluation a well-defined functional on F_μ — is exactly what fails. Every use of $\text{supp } \mu$ is therefore load-bearing for the honest-distribution case and unavailable in the setting the paper needs for its running-time consequence. Footnote 8’s worry about the ε -dependence is moot here in the strongest way: there is none.

6 Worked example

An independent check of the three identities on a case where the kernel is not an indicator. Take $n = 2$ and μ uniform on $v_1 = (1, 0)$, $v_2 = (0, 1)$, $v_3 = (1, 1)/\sqrt{2}$, $v_4 = (1, -1)/\sqrt{2}$. Then $F_\mu = \{x \in \mathbb{R}^4 : x_1 + x_2 = x_3 + x_4\}$ and $d = 3$. The kernel at $v_0 = v_1$ is $k = (3, -1, 1, 1)$, and

$$\mathbb{E}_\mu[k] = \frac{1}{4}(3 - 1 + 1 + 1) = 1, \quad K(v_0, v_0) = \|k\|^2 = \frac{1}{4}(9 + 1 + 1 + 1) = 3 = d,$$

$$\mathbb{E}_\mu[k v v^\top] = \frac{1}{4}(3v_1v_1^\top - v_2v_2^\top + v_3v_3^\top + v_4v_4^\top) = v_1v_1^\top \quad \text{exactly rank one.}$$

Normalising, $\mathbb{E}_\mu[k] = 3/2$, $c = 2/3$ and $r = (2, -\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$ with $\mathbb{E}_\mu[r] = 1$; the entropy chain is $0.14384 \leq \log \mathbb{E}_\mu[r^2] = \log \frac{4}{3} = 0.28768 \leq \log 3 = 1.09861$. Note that r is genuinely signed: it is negative at v_2 .

7 Verification record

Each round packages the artifact with the problem statement and a source card carrying every cited quotation verbatim, strips all provenance, renames it neutrally, and stages it outside the repository; five referees then read only that package, in mutually isolated fresh contexts, under a fixed referee prompt. Between rounds a handling editor rules every finding UPHELD / OVERRULED / PEDANTIC / NEEDS SOURCE, and a reviser is bounded strictly by the upheld list.

Round	Artifact	Clean	Where every substantive defect landed
1	0048-RKHS-1	1 / 5	scope commentary (Remark C)
2	r2	0 / 5	scope commentary: a false measure, $1 - o(1)$ vs 0.683
3	r3	4 / 5	scope commentary: a missing argument on the matrix domain
4	r4	2 / 5	scope commentary: the text added to repair round 3
5	r5	—	not run

Twenty passes across four rounds, over five angles — general, statement-drift and scope, measure-theoretic, line-by-line recomputation, and refutation-plus-source-fidelity — and three models, with the model rotated against the angle between rounds.

What the record shows. No pass in any round has faulted a load-bearing step. Lemmas 3–6, Theorem 1, Corollary 2 and Remark 7 are byte-identical across all five revisions, verified by hashing those regions at every boundary, and Corollary 8 is identical from r3 onward. The twelve clean readings of that text are therefore readings of the same bytes. Four passes attempted explicit refutation — on both domains, with degenerate, atomic, discrete and continuous measures — and all four failed and said so. Several independently recomputed the kernel construction, the trace identity (2), the worked example and the constant $2/(e\delta)$, and found them correct.

What keeps failing. Every substantive defect in four rounds has been in the scope commentary of §5, never in §3. The failure mode has been stable: a repair to a limiting claim reaches slightly wider than the source card supports, and the next round catches it. Round 2’s defect was a false number that a previous handling editor certified and instructed the reviser not to re-derive — which is the mechanism the multi-round structure exists to catch, working one stage later than it should have. Round 4’s defect was the text added to repair round 3. That is a prose-calibration loop, not a mathematical one, and `r5` accordingly repairs by deletion alone, adding no replacement wording.

8 Provenance, and what would make this stronger

The passes were not blind in the strict sense. The harness specifies each pass as an isolated operating-system process with an empty working directory, so that blindness is a property of the process rather than a promise. This host has no model backend — no CLI binary and no API credentials — so the passes ran instead as in-session subagents. They get genuinely fresh contexts and never see the generator’s reasoning, but they are one vendor and they run with the repository as their working directory. Mitigations applied: the package is staged outside the repository under a name unrelated to the artifact, all provenance and revision history is stripped, the referees are instructed not to read anything else, and every package is checked against a pattern list for leaked campaign machinery before dispatch. Each tally file carries this caveat in its header, and such rows must never be pooled with rows from a true `--bare` run.

Three things would materially raise confidence. First, re-running all five passes as isolated processes on a host with a backend, which is a matter of credentials rather than of work. Second, at least one pass from a different model family, which the harness recommends and which no round here has had: twenty same-vendor passes can share a blind spot that one cross-family pass would expose. Third, formalisation — the proof is short, elementary and almost entirely finite-dimensional linear algebra plus one Jensen inequality, which makes it an unusually good candidate for a proof assistant, and a machine-checked Lemma 3(b) and (2) would settle the two steps referees probed hardest.

Status of the record. Five artifacts, twenty referee reports, four editor rulings, a quotation-grounding result and the orchestrating script all sit in `c/0048/campaign/`, uncommitted. The orchestrator `scripts/verify-loop.sh` was written during this campaign because it was absent, and its first version returned the wrong termination code: it detected a blocking source defect by matching the prose “load-bearing”, a phrase the referee prompt itself contains, so referees echoing or denying it tripped the test. It now tests the findings table structurally. That script has not itself been reviewed.